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12CE32/42

III SEMESTER ENGINEERING EXAMINATIONS DECEMBER 2013

Ecology and Engineering Environment

Common to ALL Branches

Duration: 2 Hours

Max Marks:*40*

Instructions: **1. Answer all the Questions, each question carries ONE MARK.**
2. Indicate answer in separate answer sheet provided.
3.Damaging/overwriting, using whitener on the answer sheets are strictly prohibited.

1. Which one of the following pesticides is banned now-a-days?

a. Isoprene
b. DDT
c. Endrin
d. Aldrin

2. In metropolitan cities main source of pollution is:

a. Leakage of gas from factories
b. LPG cooking gas
c. Smoke of coal
d. Exhaust of automobiles

3. Acid rains occur due to:

a) Presence of CO_2
b) Presence of CO and CO_2
c) Presence of N_2
d) Presence of SO_2 and SO_3

4. Sewage drained into water bodies kill fishes because:

a. Excessive CO_2 is added in water
b. It gives off a bad smell
c. It removes the competition among fishes to dissolved CO_2
d. It increases the competition among fishes to dissolved oxygen

5. Electrostatic precipitator is for:

a. Control of water pollution
b. Control of air pollution
c. Both of these
d. None of these

6. The study of interactions between living organism and environment is called as

a. Ecosystem
b. Ecology
c. Phytogeography
d. Phytosociology

7. The term ecology was introduced by

a. Haekel
b. Odum
c. Tansely
d. Ramdeo Mishra

8. The best soil for healthy and vigorous growth of most plants is

a. Clayey
b. Sandy
c. Loamy
d. Clayey loam

9. The annual productivity of any ecosystem is greater than the annual increase in biomass of the herbivores in the ecosystem because.

a. plants convert energy input into biomass more efficiently than animals
b. there are always more animals than plants in any ecosystem
c. plants have a greater longevity than animals
d. during each energy transformation, some energy is lost
e. animals convert energy input into biomass more efficiently than plants do

10. Food chain in which microorganisms breakdown the food formed by primary producers is

a. Parasitic food chain
b. Detritus food chain
c. Consumer food chain
d. Predator food chain

11. "Minamata" disease is produced due to the pollution of water with:

a. Silver nitrate
b. Mercuric chloride
c. Methyl isocyanate
d. Crude oil

12. Eutrophication results in the reduction of:

a. Mineral salts
b. Dissolved hydrogen
c. Dissolved oxygen
d. None of these

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13. The pollutants released by jet planes are:

a. Aerosols
b. Colloids
c. Smog
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17. Which of the following is the most important natural resource of energy?

a. Electricity
b. Biogas
c. Fossil fuels
d. Atomic fission

18. Atmosphere absorbs maximum amount of:

a. Visible light
b. Near ultraviolet light
c. Far ultraviolet light
d. Infra-red waves

19. The yellow-green color of coastal waters as compared to the blue color of the open ocean is a result of what?

a. A lower salinity
b. A high concentration of organic materials
c. A higher copper content of sediment
d. A low copper content of sediment

20. What type of shore is most biologically diverse?

a. Sandy beach
b. Mud flat
c. Salt marsh
d. Rocky shore

21. What is the depth below which net primary growth cannot occur?
a. Dissolved oxygen minimum
b. Biological compensation depth
c. Critical depth
d. Oxygen compensation depth

22. The oxygen minimum zone is partially a result of water with
a. High temperature
b. High productivity
c. Low productivity
d. Low temperature

23. What is the type of exoskeleton formed by corals?
a. Hydrogen Calcite
b. Calcium Carbonate
c. Limestone
d. Phosphate

24. The water cycle is made up of four processes that work together. Which of the following is correct ?

a. Evaporation, Transpiration, Condensation & Precipitation
b. The liquid becomes frozen and becomes ice.
c. The sun dries up the water in the clouds.
d. The water stays in the clouds

25. Which of the following radiations is non- ionizing and has more specific biological effect than others?

a. X-rays
b. beta rays
c. gamma rays
d. Ultraviolet rays

26. What is the best definition of a reservoir?

a. A man made lake used to store water for community use
b. A place where community residents can swim and bathe
c. A structure where water is cleaned
d. A tool to measure the amount of rainfall in a community

27. As time goes by, and water goes through the water cycle again and again, the amount of water on Earth

a. Increases
b. Decreases
c. Stays the same
d. Goes up and down

28. Snow and hail are examples of
a. Runoff
b. Precipitation
c. Condensation
d. Water vapor

29. Red soils of India are:

a. Laterite soils
b. Deficient in lime, Mg, P and K
c. Rich in iron and organic matter
d. All of the above

30. When biosphere turns into human dominated environment, it is called:

a. Stratosphere
b. Troposphere
c. Mesosphere
d. Thermosphere

31. Photochemical smog is due to:

a. Rapid increase in concentration of $C0_2$
b. Presence of high humidity in atmosphere
c. Depletion of ozone layer
d. Rapid builds up of dust particles, pollutants and thermal inversion

32. Which disease in children is caused by intensive use of nitrate fertilizer?

a. Mumps
b. Septicemia
c. Methemoglobinemia
d. Jaundice

33. The noise pollution is measured in:

a. Decibel (dB)
b. Fathoms
c. Tonnes
d. Kilograms (kg)

34. Large volcanic eruptions appear to make the Earth's climate _____, at least temporarily.
a. drier
b. more stormy
c. wetter
d. cooler
e. less stormy

35. In addition to their role in ozone depletion, CFCs play a role in global warming by:
a. Absorbing solar radiation
b. absorbing earthshine in the 10 micron region
c. blocking ultraviolet-B radiation
d. destroying ozone, which cools the Earth

e. reducing the albedo of the Earth

36. In Precambrian time, oxygen was first produced by primitive algae and bacteria. For them, the oxygen was:

a. necessary for respiration
b. toxic
c. necessary for photosynthesis
d. irrelevant
e. a product of chemosynthesis

37. With an increase in altitude, the average change in temperature in the Earth's troposphere is:

a. 6.5°F warmer per km
b. 6.5°F cooler per km
c. 6.5°C cooler per km
d. called the adiabatic lapse rate
e. impossible to measure

38. Why does only approximately half of the solar energy emitted towards the Earth ever arrive at the surface?

a. the rest is lost in transit through the vacuum of space
b. the rest is lost because the speed of the light is slower in the atmosphere than in the vacuum
c. the rest is absorbed by the atmosphere or reflected into space
d. the rest is refracted through the atmosphere
e. the rest is converted into heat energy

39. How does air temperature change with altitude in the upper stratosphere and why?

a. it decreases due to the presence of water vapor
b. it decreases or increases, depending on the time of day
c. it increases, due to the destruction of ozone by ultraviolet light
d. it increases due to the greenhouse effect
e.it is isothermal, due to loss of earthshine to space

40. Incoming solar radiation is partly reflected back into space by the Earth's:

I. atmosphere
II. land surface
III. ocean surface

a. I only
b. II only
c. III only
d. II and III
e. I, II, and III